

AMD RYZEN 7 8700G PERFORMANCE

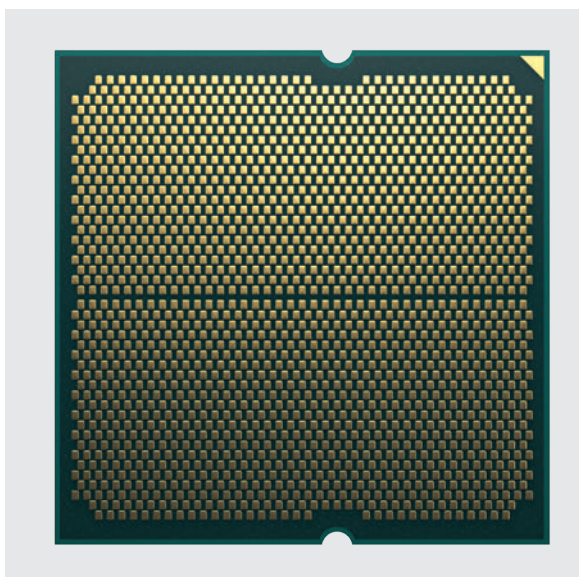
Like always, we begin with Cinebench and the 7800X3D is about 600 points ahead of the 8700G in multithreaded runs and by about 80 odd points in the single-threaded run. In UL Procyon, the 8700G performs equal to the 7950X in the Word benchmark but in Excel, PowerPoint and Outlook, it's a little better than the 5600G and behind the 7800X3D. So for home and office use, this is a pretty decent processor. If you like to play around with a lot of scripts and macros in Excel, then you might want to explore the 7800X3D instead.

For video editing workloads, the 8700G is quite evenly matched against the 7800X3D. That being said, the Intel 14600K is actually a much better processor in this benchmark. In Blender, we see the 7800X3D outperform the 8700G and the Intel 14600K outperforms the 7800X3D.

The 8700G doesn't consume a lot of power. In fact, it is very much in line with the older 5600G and the newer 7800X3D at about 86-88 Watts at peak load. Since these parts were initially designed for mobile form factors such as thin-and-light laptops, there are a lot of temperature and power throttling mechanisms in place to manage battery life. While some of these limitations are removed for desktop parts, the base design considerations still result in lower power consumption. .

GAMING ON DISCRETE GRAPHICS

Since the IGP is the centre of attention, we've split the gaming performance section to focus on discrete GPU and



integrated GPU. Starting off with the discrete GPU, the 7800X3D continues to rule the roost. It easily outperforms processors that are twice as expensive and the 8700G with its reduced L3 Cache does lag behind. The 7700X, 7600X, and the Intel 14600K are all ahead of the 8700G.

GAMING ON INTEGRATED GRAPHICS

The RDNA 3-based IGP on the AMD Ryzen 7 8700G is miles ahead of its predecessors. The IGPs on the 5600G or the older-gen Intel processors cannot hold a candle to the 8700G. In

games such as Far Cry 6, the 8700G was easily knocking out more than 85 FPS on Low settings and in GTA V, it was giving more than 130 FPS. And in Shadow of the Tomb Raider, you would see about 66 FPS on average. All around great numbers for an IGP.

That being said, an entry-level card such as the GTX 1650 which costs about INR 13K will still perform better than the 8700G. So, if you're an eSports player getting into CS2 or Valorant, then the 8700G is an excellent processor but most gamers aren't going to stick to just these two video games. You'd want to try out more titles and that's where a discrete GPU becomes essential.

VERDICT

If developed markets, the 8700G would be termed a decent buy for a budget gaming PC. If fact, I'd consider getting the 8600G if saving money was a priority. But we have the 7800X3D for just a few thousand rupees more and that makes a lot more sense for a gaming machine. It also has an IGP, albeit not as powerful, so one could opt for the X3D processor if you're looking to get a stop-gap until a discrete GPU can be purchased. When we benchmarked the CPU we weren't aware of the STAPM (Skin Temperature-Aware Power Management) issue which has been disabled. That gives the 8700G a teensy bit more performance but we'd continue to recommend the 7800X3D just because it's better in more aspects than the 8700G.

—Mithun Mohandas

SPECIFICATIONS

Cores: 8. | Threads: 16 | GPU Cores: 12. | Base clock: 4.2 GHz | Boost clock: 5.1 GHz | Process node: 4nm | L3 Cache: 16 MB | L2 Cache: 1 MB per core | TDP: 65 W.

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